

SRITIZ SAHU

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PROFESSIONAL SUMMARY

Instrumentation and Control Engineering student at NIT Jalandhar with a Minor in Computer Science. Specialized in **Full-Stack Development (MERN, FastAPI)** and **Applied Machine Learning**. Proven experience in architecting scalable backend controllers for university-wide platforms and implementing advanced **Computer Vision** and **Explainable AI (XAI)** frameworks to solve complex software challenges.

EDUCATION

Dr. B.R. Ambedkar National Institute of Technology (NIT Jalandhar) Expected: May 2028
Bachelor of Technology in Instrumentation and Control Engineering (CGPA: 8.00/10.0) Jalandhar, Punjab
Minor in Computer Science and Engineering

TECHNICAL SKILLS

- **Languages:** C++, Python, JavaScript (ES6+), SQL, C, HTML/CSS
- **Web Development:** MERN Stack (MongoDB, Express, React, Node.js), FastAPI, RESTful APIs, ShadCN UI, Tailwind CSS, Context API, Redux, JWT Authentication
- **Machine Learning/AI:** Large Language Models (LLMs), Generative AI, Explainable AI (SHAP), TensorFlow, Keras, Scikit-learn, Computer Vision, OCR (Tesseract), Pandas, NumPy
- **Tools & Infrastructure:** Git/GitHub, MongoDB Atlas, Firebase/Firestore, Postman, Render, Vercel, Google Colab

EXPERIENCE

Indian Institute of Technology (IIT Patna) Dec 2025 – Jan 2026
Research Intern (Supervisor: Prof. Maheshkumar H. Kolekar) Patna, Bihar

- Authored technical content and architectural diagrams for a book chapter on **Generative AI and VAEs** in Video Surveillance, synthesized for publication.
- Performed architectural analysis of state-of-the-art Computer Vision papers, including **Attention-Based UNet** and **MDPNet** (Multiscale Dynamic Polyp-Focus Network).
- Evaluated medical image segmentation methodologies, identifying technical bottlenecks to streamline research direction.
- Received a formal invitation for future summer projects based on analytical depth and technical contributions.

PROJECTS

SmartDocs: AI-Powered File Organizer | MERN Stack, OCR, MongoDB GridFS Jan 2026

- Built a full-stack MERN application that automates document organization using rule-based heuristics and fallback **Machine Learning** models.
- Implemented an **OCR pipeline** using OCRmyPDF and Tesseract.js to extract text from scanned PDFs, significantly improving classification accuracy for low-text images.
- Designed a scalable storage system using **MongoDB GridFS** for efficient large file handling.

TrustGuard: AI Fraud Detection Platform | FastAPI, LLMs, React, Scikit-learn, SHAP Feb 2026

- Developed an intelligent healthcare claims platform featuring **LLM-based document analysis** to detect anomalies in clinical text and structured billing data.
- Integrated **Explainable AI (SHAP)** to provide transparent reasoning for fraud scores via waterfall visualizations, utilizing a Logistic Regression pipeline for deterministic risk assessment.
- Built an interactive dashboard using **Recharts** for risk-gauge visualization and automated PDF report generation.

Brain Tumor Detection Using Deep Learning | Python, TensorFlow, Keras, OpenCV Oct 2025

- Implemented a Deep Learning solution using **Transfer Learning (VGG16)** to classify tumors from brain MRI scans.

- Engineered custom Python generators to manage 2GB+ of image data for memory-efficient training, achieving a 15% improvement in classification confidence scores.

LEADERSHIP / EXTRACURRICULAR

TPO Website Team | Website Development Team Member

Aug 2025 – Present

- **Engineered the Faculty Dashboard from scratch**, implementing full-stack logic to manage placement data.
- Collaborating on the placement portal used by recruiters from **100+ companies** and graduating seniors.

Xceed (Official NITJ Web Dev Club) | Club Member

Aug 2025 – Present

- Developed the "VISTA ECE" department website and **architected backend controllers** for university timetable systems.
- Maintained university platforms used by **3,000+ students**, ensuring 99% uptime during high-traffic periods.